Machine Learning Analysis of Multimodal Health Dataset

Predict High vs. Low Loneliness

Intelligent Systems - CSC 500
Zoe Rochelle
Team 3
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0 Workflow tools

Tech Stack

- Workspace: VS Code/Github
- Platform: AWS
  - Storage: S3
  - Compute: Glue/Spark
  - SQL/DB: Athena

Modules/Libraries

- PySpark (ETL)
- Pandas
- NumPy
- Scikit-Learn
- XGBoost

0 Workflow overview: AWS

S3 Glue Athena

S3 -> store the dataset and tables

Glue/Spark -> data cleanup, transformations, building Athena tables, data processing

Athena -> querying/viewing data: Aware, Oura, Watch, Survey

1 Problem Definition

- Goal: predict **high vs. low loneliness** from a multimodal health dataset.
- This is defined as **supervised learning**, and a binary classification problem.
- Target **labels** were created from the 20-item UCLA Loneliness Scale.
- Each participant received one total UCLA score after response mapping and reverse scoring: Never -> 1, Rarely -> 2, etc.
- A median split was used: above the median = high loneliness; at or below the median = low loneliness.
- Each row in the final modeling dataset represents one participant.

2 Data Preprocessing

- The dataset included **AWARE** smartphone data, **Oura** wearable data, and **survey** data.
- Raw CSV files were cleaned, renamed when needed, and converted into Parquet datasets.
- Schemas and Athena tables were checked and corrected so the data types matched the Parquet files – had some issues here.
- Survey data was reshaped into a long-format table for easier filtering and scoring.

2 Data Preprocessing (cont.)

The survey files are inconsistent: different surveys can have different numbers of questions.

```
## Wide format ##
participant     q1          q2          q3
Participant_1   Often       Rarely      Never
Participant_2   Sometimes   Often       Rarely
```

This change made the survey files easier to query.

```
## Long format ##
participant     question_id   response_raw
Participant_1   q1            Often
Participant_1   q2            Rarely
Participant_1   q3            Never
Participant_2   q1            Sometimes
Participant_2   q2            Often
Participant_2   q3            Rarely
```

Now we can filter rows like:

- only UCLA surveys
- only q1
- only one participant

2 Data Preprocessing (cont.)

- UCLA responses were mapped from text to numbers, reverse-scored for positive items, and summed into a total loneliness score. Positively worded questions were **manually identified**.

  Higher average = more loneliness
  Lower average = less loneliness

  I feel lonely: 4
  I feel connected: 4 (becomes) 1

  So, when we average up the scores, happy responses lead to a lower score.

- Participant-level features were created from AWARE and Oura tables, and missing numeric values were filled with the median.

3 Models Used

- Four models were trained, as required by the project: SVM, Random Forest, XGBoost, and a Neural Network.
- The **input feature** was built from participant level AWARE and Oura features.
- The **target variable** was loneliness_label_binary.
- As per the instructions, I decided to use **stratified cross-validation** as a way of splitting the data into several training/testing folds. Based on my research, this would keep the evaluation fairer and more consistent on a small dataset.
- Also, as per the instructions, the models were evaluated using **accuracy**, **precision**, **recall**, **F1 score**, and **ROC-AUC** (measures a model's ability to distinguish between classes).

3 Models Used (cont.)

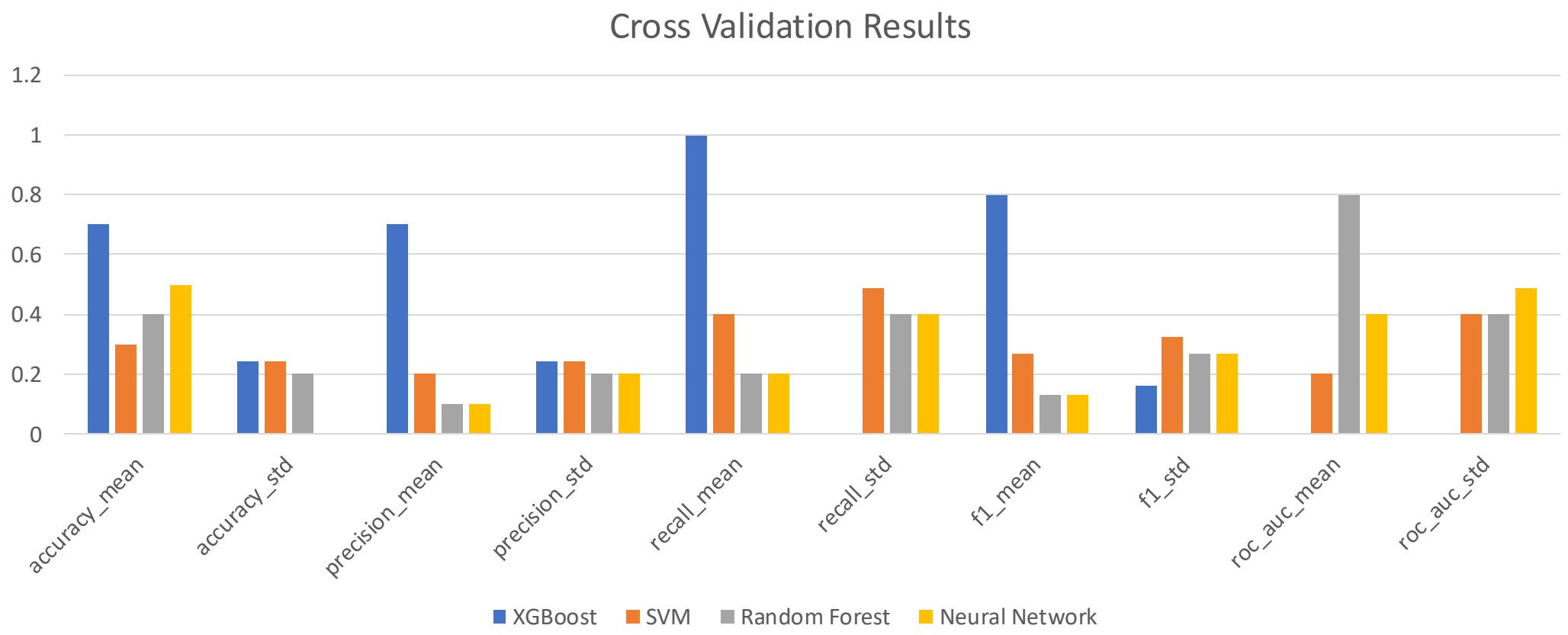
Also, as per the instructions, the models were evaluated for:

Accuracy: more predictions are correct
Precision: when the model predicts high loneliness, it is correct more often
Recall: what percentage of actual high-loneliness participants did the model find
F1 Score: better balance between precision and recall
ROC-AUC: how well the model seperates between the two classes

4 Results and Insights

- **XGBoost** performed best overall and achieved the strongest prediction results.
- **Random Forest** did not perform as well, but it provided a more balanced and interpretable feature-importance profile.
- **Important features** included battery related phone data, sleep efficiency, awake time, deep sleep, resting heart rate, and call duration.

4 Results and Insights (cont.)



Higher = better

5 Conclusion

- These findings suggest that loneliness may influence both phone usage behavior and sleep or recovery patterns.
- In this project, **battery related** phone features were especially important, along with sleep related features like awake time, sleep efficiency, deep sleep, sleep duration, and resting heart rate.
- This suggests that loneliness may affect daily habits, communication, and some physical routines.
- However, because the dataset is small overfitting is a possible issue, and because XGBoost relied heavily on one feature.
- Using **stratified cross-validation** helps, but doesn't eliminate the issue completely. Interpret these results with caution.

6 Sources

Source paper
Smartphone Data Gathered Early in Depression Treatment Predicts Treatment Outcome - https://dl.acm.org/doi/10.1145/3721201.3721380

Source dataset
Loneliness and well-being in Finnish immigrants: A multimodal dataset from wearables and passive data collection
https://datadryad.org/dataset/doi:10.5061/dryad.qz612jmrn

Models
https://scikit-learn.org/stable/modules/svm.html
https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.RandomForestClassifier.html
https://scikit-learn.org/stable/modules/neural_networks_supervised.html
https://xgboost.readthedocs.io/en/stable/get_started.html